Nachum Getzel Elkind

Software Engineer | Low-Level Systems | Cybersecurity

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> *An exceptional talent with rare deep-systems intuition. Nachum combines the theoretical rigor of a top-tier academic with the practical 'hacker' mindset needed to solve complex low-level challenges.*
>
> *— Inbar Levy, Magshimim course instructor*

Summary

High-achieving software engineer with advanced expertise in low-level systems, algorithm design, and cybersecurity. Graduate of Israel's elite Magshimim National Cyber Program with honors, combining deep theoretical foundations in Computer Science with hands-on experience in reverse engineering, assembly, and systems architecture. Proven track record in national-level competitions and university-level academic excellence while in high school.

Technical Expertise

Low-Level Systems & Security

- Assembly (8086) programming and architecture
- Reverse engineering and binary analysis (IDA Pro)
- Network protocol analysis and packet inspection (Wireshark, Fiddler)
- OS fundamentals and memory management

Software Development

- AI-assisted development (ChatGPT, Grok, Claude, Gemini) for code generation and optimization
- Coding agents (Codex, Cursor)
- Object-Oriented Programming (OOP) design patterns
- Algorithm design and complexity analysis
- Client-Server architecture and socket programming
- System integration and API development

Languages & Tools

- Languages: C, C++, Assembly (8086), Python, Java, JavaScript, Node.JS
- Tools: Git, GitHub, Wireshark, IDA Pro, Fiddler
- Platforms: Windows, Linux

Languages Spoken

- Russian (C2 - Native/Bilingual)
- Hebrew (C1 - Professional Working Proficiency)
- English (B2 - Professional Working Proficiency)

Projects

Chess Engine | C++, OOP

Architected and developed a full-featured chess engine from scratch. - Implemented complex game logic using advanced Object-Oriented design patterns. - Optimized move generation and board evaluation algorithms for performance. - Designed robust memory management system for game state handling.

Client-Server Application | Python, Sockets, Networking

Built a robust distributed system using raw socket programming. - Implemented custom application-level protocols on top of TCP/IP. - Handled concurrency and multiple client connections efficiently. - Designed secure message passing and error handling mechanisms.

Telegram Bot Integration | Node.JS, OpenAI API, Google Calendar API

Developed an intelligent scheduling assistant integrating multiple external APIs. - Integrated ChatGPT for natural language processing of user requests. - Implemented real-time two-way sync with Google Calendar. - Built scalable backend infrastructure in Node.js to handle webhook events.

Chrome Extension | JavaScript, ChatGPT API

Created a browser-based productivity tool leveraging AI. - Engineered seamless DOM manipulation for on-page content interaction. - Integrated asynchronous API calls for real-time AI responses.

Supermarket Price Scraper | Python, Web Scraping

Developed an automated data collection tool for retail analytics. - Reverse-engineered web requests to bypass anti-scraping measures. - Implemented data normalization and structured storage for analysis.

Education

Magshimim National Cyber Program, Israel National Cyber Directorate, 2021-Present

- Elite cyber training program for gifted students
  - First Year Average: 97.5
  - Second Year Average: 99.7
  - Focus: Algorithms, Low-level programming, Networks, Cyber warfare
  - Participant in national-level capture-the-flag (CTF) competition stage

High School Diploma, High School, 2027 (Expected)

- Excellence Track
  - Overall Average: 95+
  - Computer Science (5 units): 100

Academy in High School, The Open University of Israel, 2023-Present

- Early Academic Track
  - Accepted to B.Sc. Computer Science track during high school
  - Successfully completed first-year mathematics-oriented course